

SHANTANU SARKAR

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PROFESSIONAL SUMMARY

Motivated BCA student and Web Developer with hands-on experience in building responsive, user-centric web applications using HTML, CSS, JavaScript, and Python (Flask). Experienced in frontend development, backend fundamentals, and database integration, with a growing focus on Data Structures & Algorithms and DevOps fundamentals. Passionate about continuous learning and building scalable, real-world software solutions.

EDUCATION

Gandhi National College (Utkal University)

Bachelor of Computer Application (BCA)

Bhubaneswar, Odisha

2021 – 2026

- Focusing on Core Computer Science principles including Data Structures, DBMS, and Software Engineering.

Dhanbad Public School

Class XII (Science Stream)

Dhanbad, JH

2021 – 2023

EXPERIENCE

Web Development Intern

Bharat Coking Coal Limited (BCCL)

June 2025 – July 2025

Dhanbad, JH

- Designed and developed a responsive corporate website using HTML5, CSS3, and JavaScript, ensuring a 20% increase in cross-device accessibility.
- Implemented mobile-first design strategies and verified performance across multiple browser engines.
- Revamped internal navigation systems to improve user flow and reduce bounce rates on key pages.
- Applied frontend optimization techniques such as asset minification and lazy loading for faster load times.
- Collaborated with senior IT staff to align web components with corporate branding and security guidelines.
- Documented technical workflows for deployment and ongoing maintenance of the site modules.

PROJECTS

StudyPilot AI | Python, Flask, JavaScript, SQLite, OpenAI API

2024

- Architected a full-stack AI-powered study planner application utilizing frontend deployment on Netlify and backend services on Render.
- Integrated the OpenAI GPT API to develop an intelligent chat assistant for subject-specific academic queries.
- Developed automated scheduling algorithms and progress-tracking features with a responsive dark-mode interface.

AES File Encryption Tool | Python, AES-256-GCM, Flask, Cryptography

2024

- Developed a secure file encryption utility utilizing the military-grade AES-256-GCM algorithm with PBKDF2 key derivation.
- Engineered a modular architecture supporting both a standalone desktop GUI and a local web-based browser interface.
- Implemented robust file-handling logic with integrity verification to protect sensitive user data from tampering.

ScamShield | Python, Flask, HTML/CSS/JS, SQLite

2024

- Designed and built a security-focused web application featuring a pattern recognition system to detect and report online scams.
- Created a real-time risk assessment engine that evaluates user-submitted URLs and displays detailed safety scores.
- Authored a 70+ page formal academic project report encompassing system design, cost estimation, and flowcharts.

Web-Based Quiz Application | Python, Flask, HTML, CSS, JS, SQLite

2024

- Architected a full-stack quiz platform featuring secure user authentication and dynamic question generation.
- Developed robust backend logic using Flask and SQLite to handle session-based scoring and user profiles.
- Implemented an intuitive MVC architecture to separate business logic from the responsive UI layers.

Personal Portfolio Website | HTML5, CSS3, JavaScript

2024

- Built a high-performance personal brand site utilizing modern CSS Grid and Flexbox for fluid layouts.
- Enhanced interactivity through custom JavaScript animations and smooth-scrolling components.
- Integrated the GitHub API to dynamically fetch and display recent public repositories on the homepage.

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML5, CSS3, SQL (MySQL, SQLite)

Frameworks: Flask

Developer Tools: Git, GitHub, VS Code, Vercel, Netlify, Render, Linux/Terminal

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), RESTful APIs, Responsive Web Design, Cryptography Basics

SOFT SKILLS

Interpersonal: Problem Solving, Team Collaboration, Effective Communication, Adaptability

Operational: Project Coordination, Time Management, Attention to Detail, Continuous Learning